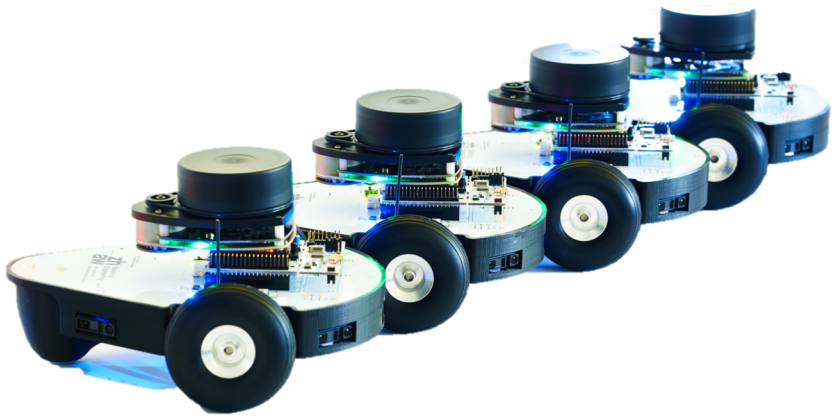
Zurich University
of Applied Sciences
School of
Engineering
IMS Institute of
Mechatronic Systems

# Einführung Elektronik (PES-Board)

Toby Rotach, FS 2018 SW 2



# PES 2 Board – Funktionen

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- Eine IMU (Inertial Measurement Unit mit Gyro, Beschleunigungssensoren und Magnetometer)
- Drei Motorentreiber für die direkte Ansteuerung von DC-Motoren (Nennstrom 1A)
- Drei Encoder-Counter um die Umdrehungszahlen einzulesen
- Vier digitale I/O, 3.3V (5V tolerant)
- Vier analoge Eingänge, 3.3V (5V tolerant)
- High Side Switch, 5V (Ausgangsstrom max. 1A)
- Stecker um die Motoren, Encoder, Sensoren anzuschliessen
- 12V NiMH Akkupak 2300mAh
- Ladebuchse und einfache Spannungsanzeige für den Akku

# PES 2 Board - Komponenten

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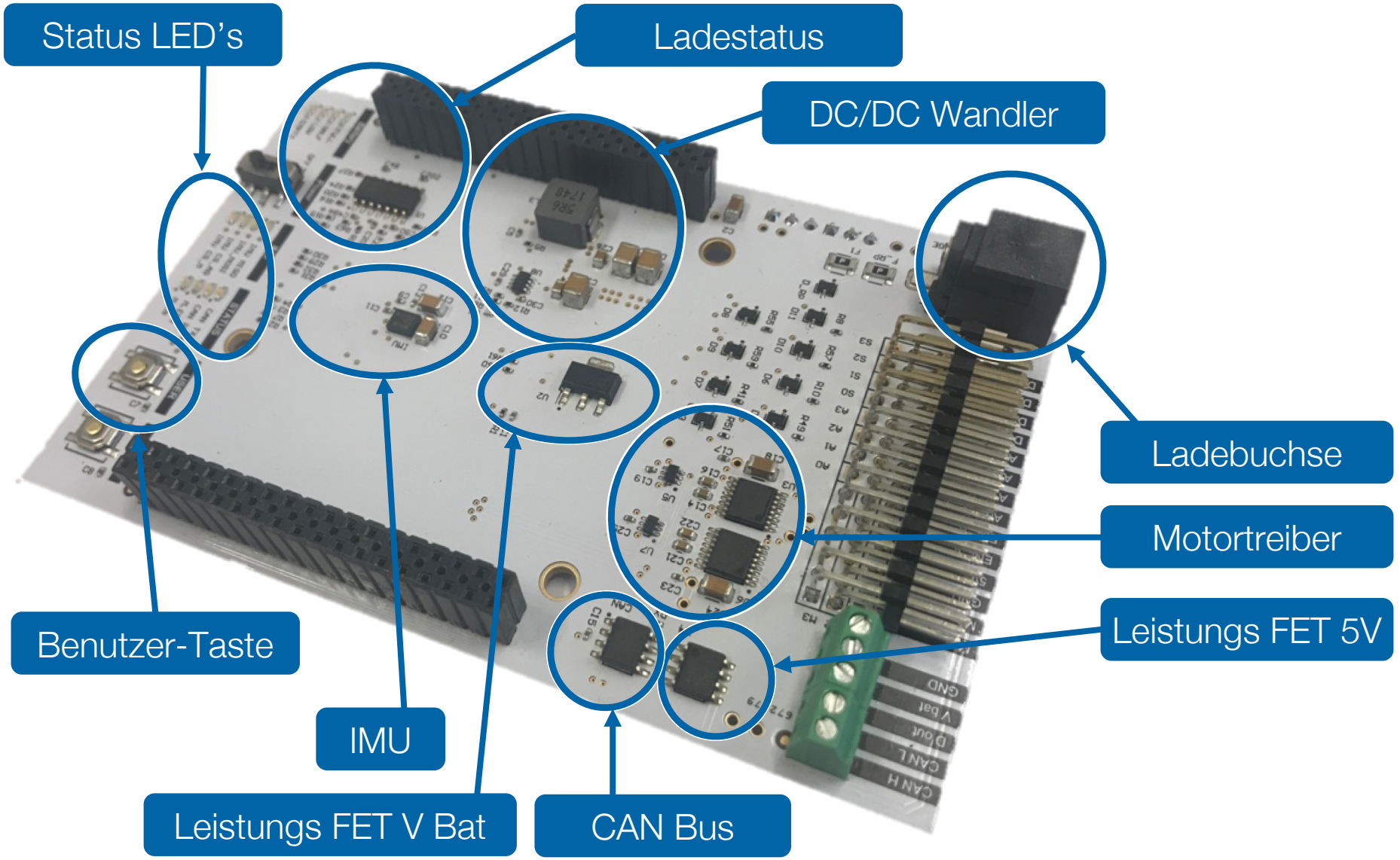

# PES 2 Board – Layout Top Layer

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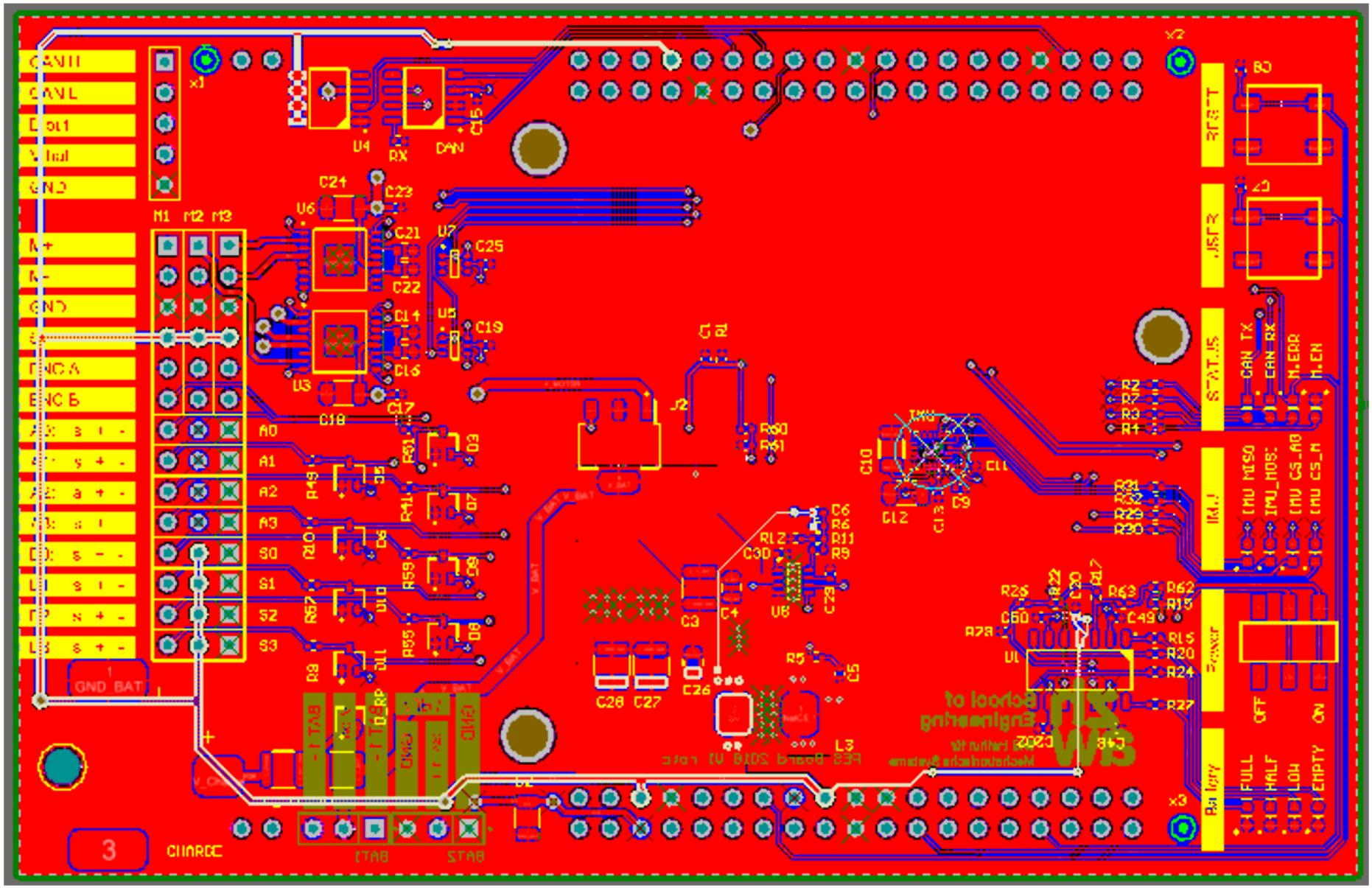

# PES 2 Board – Layout Bottom Layer

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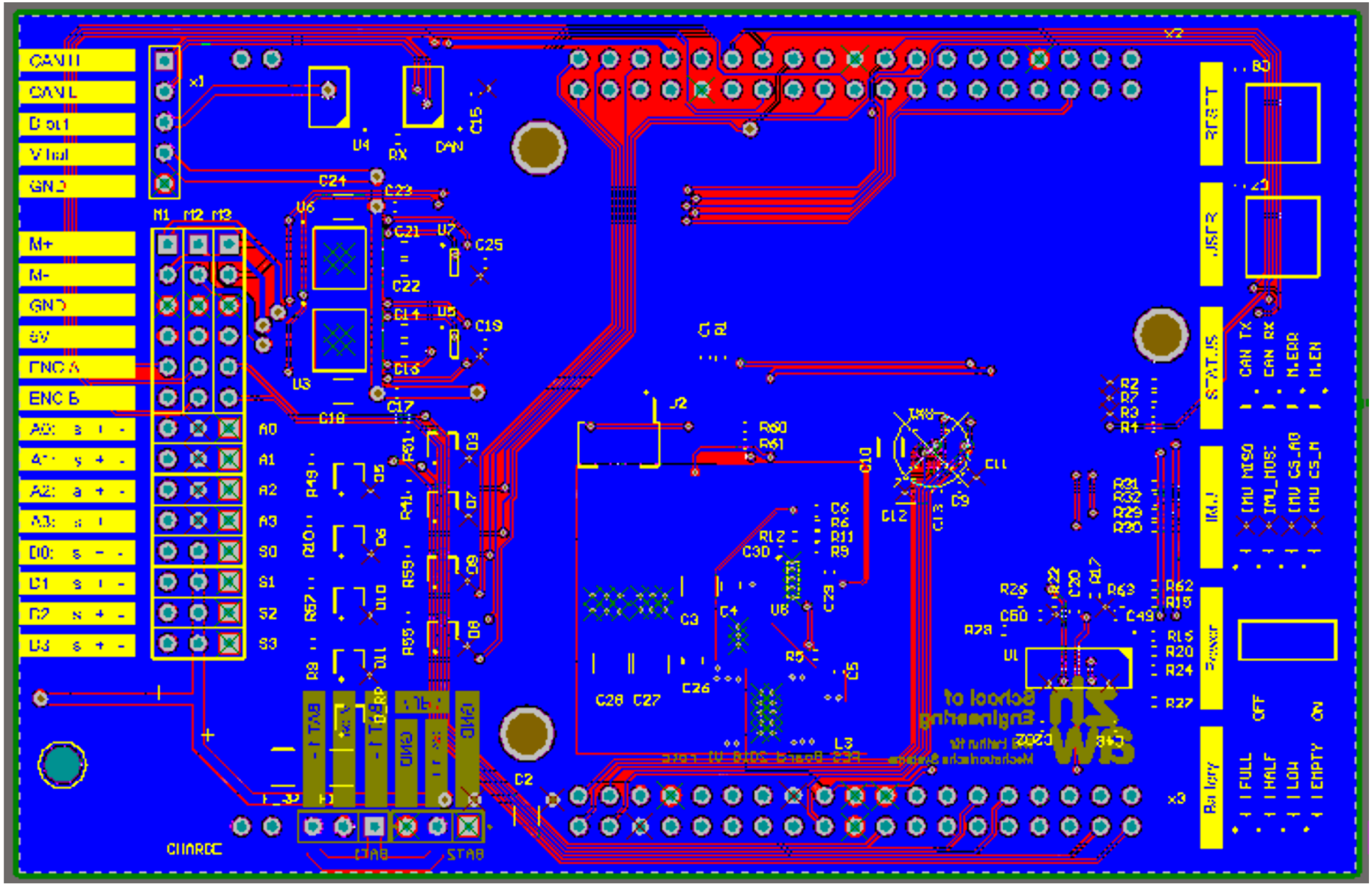
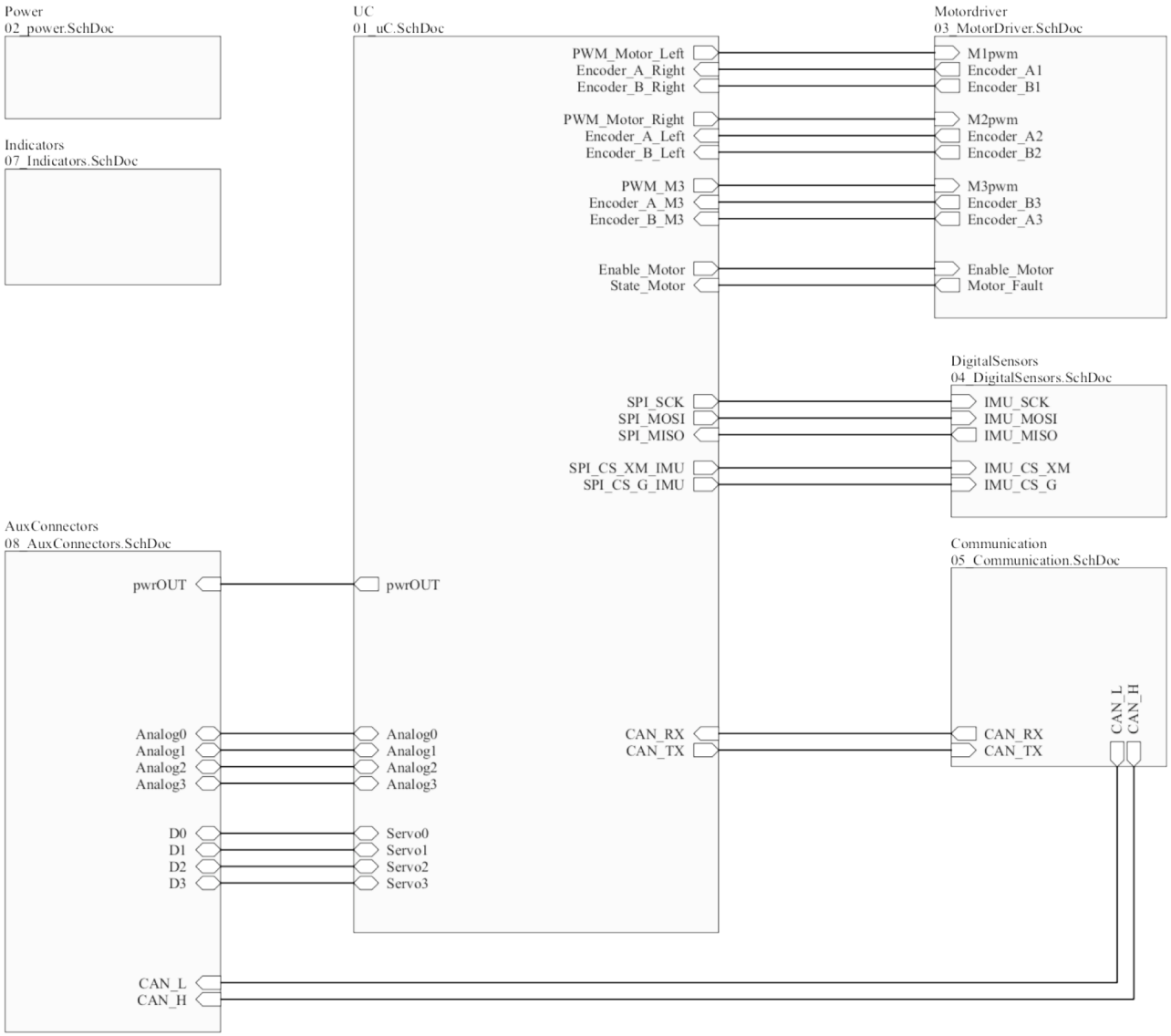
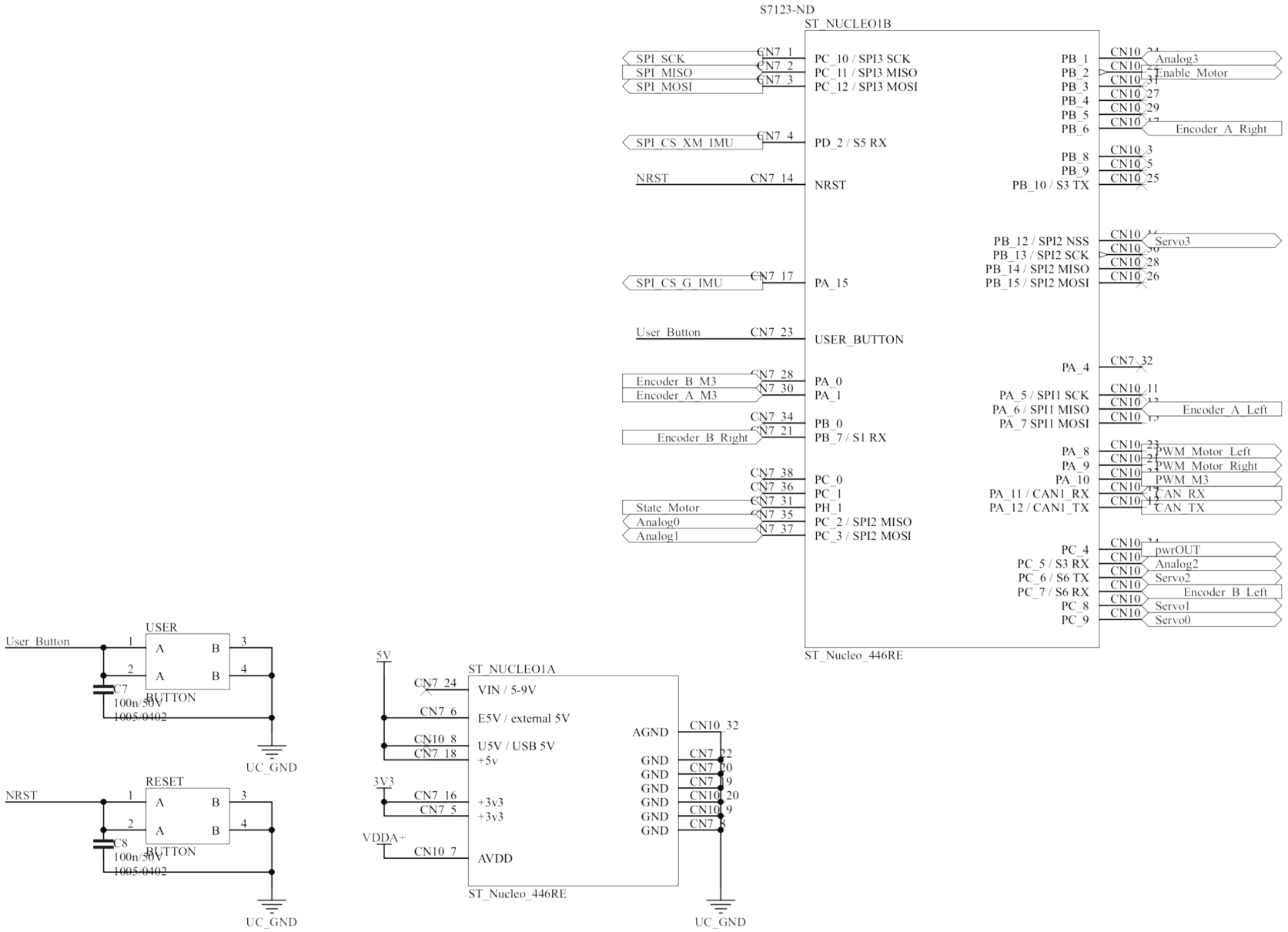


# PES 2 Board - Baugruppen



# PES 2 Board – uC Anbindung



# PES 2 Board - Peripherie

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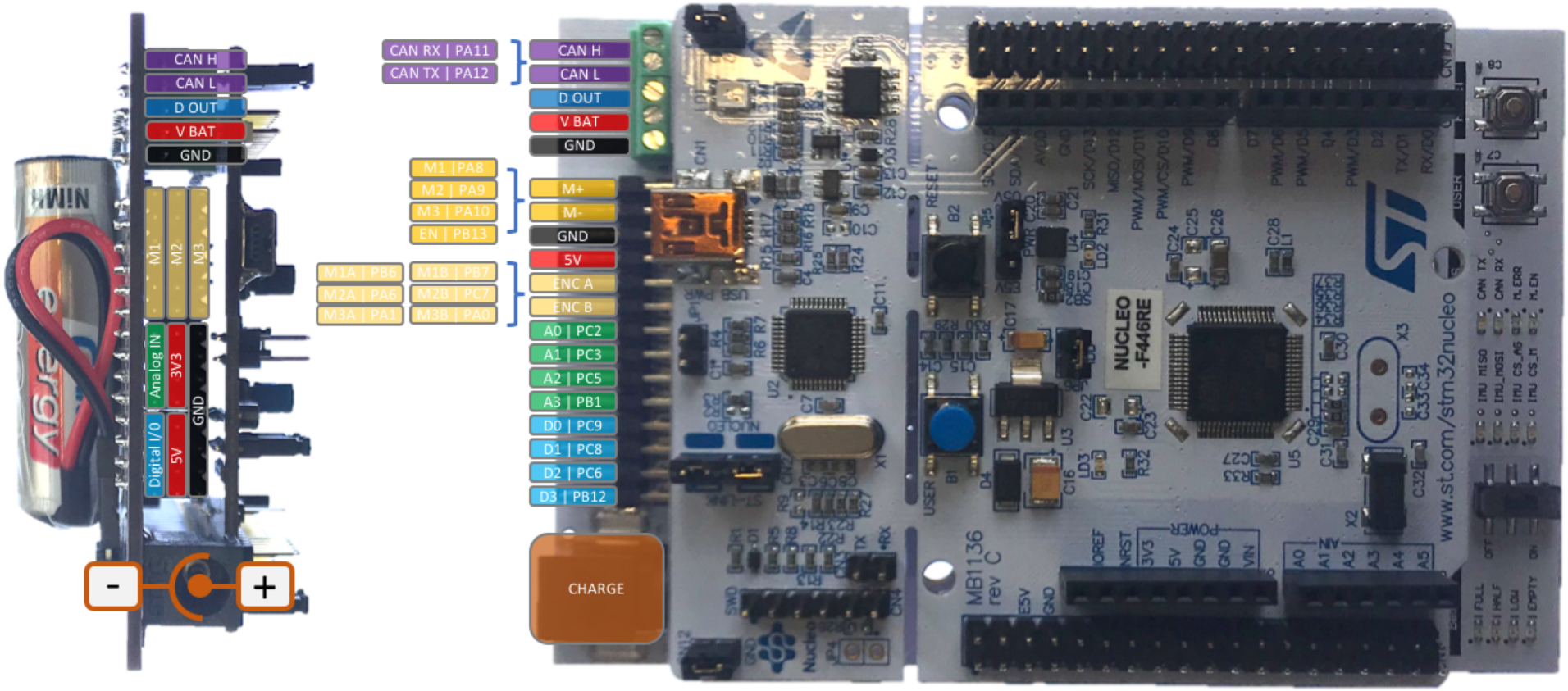
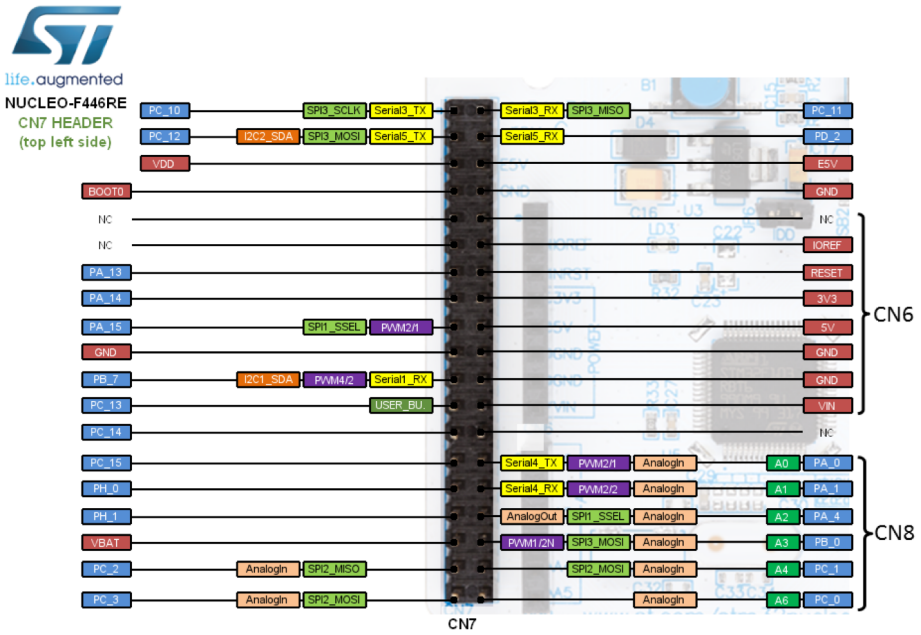
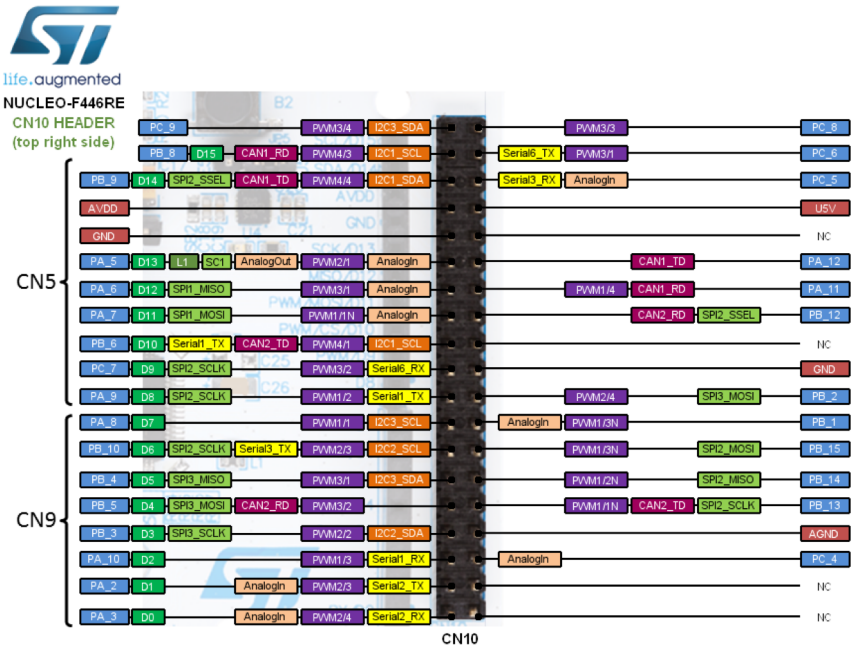


# PES 2 Board – ST Morpho Headers



# PES 2 Board – Arduino Headers

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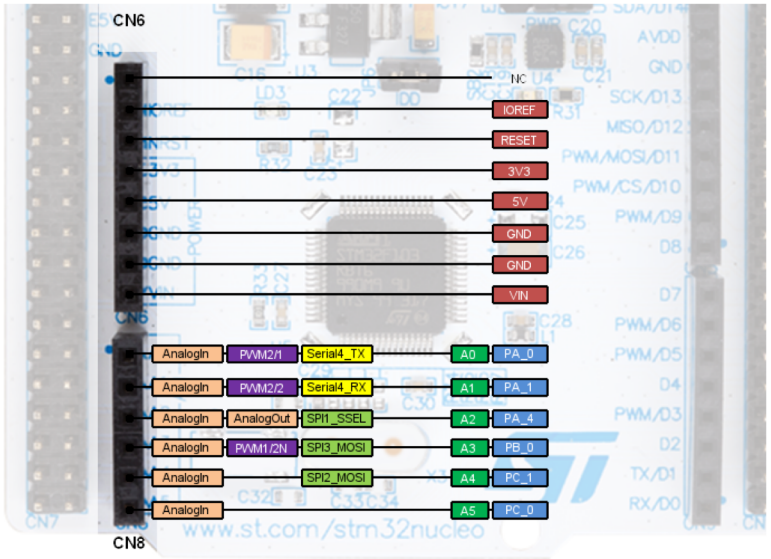
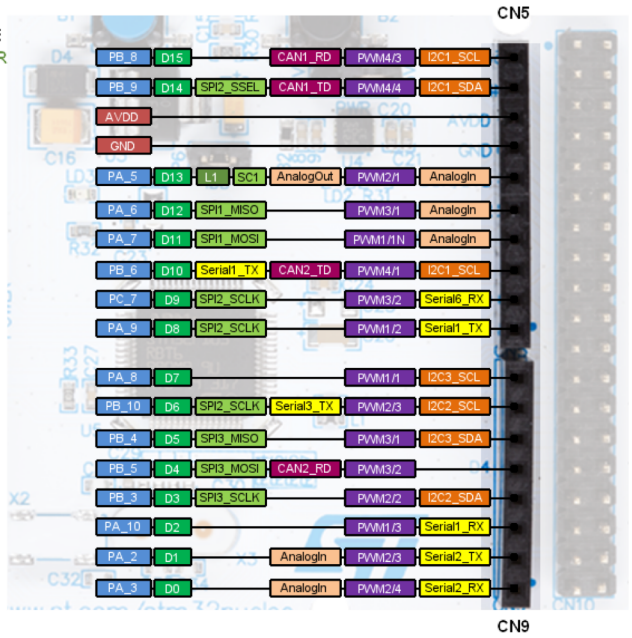
**NUCLEO-F446RE**
ARDUINO HEADER
(top left side)




life.augmented

**NUCLEO-F446RE**
ARDUINO HEADER
(top right side)



# PES 2 Board – Pinmap komplett

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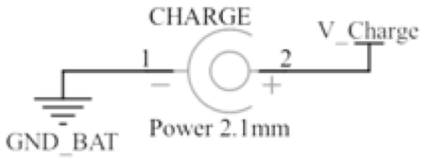
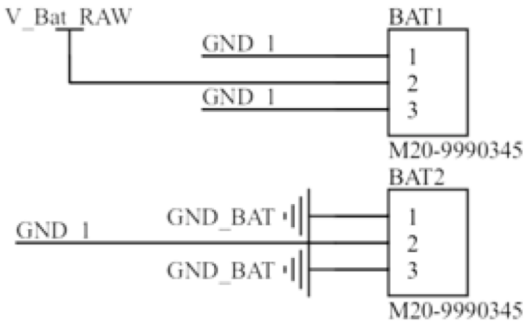
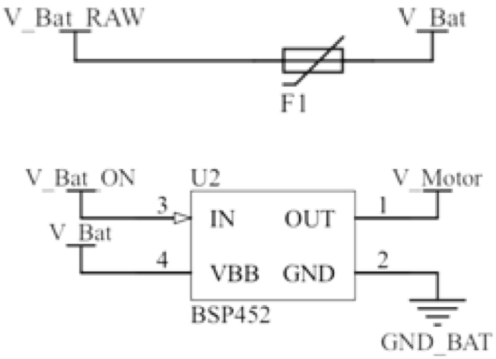
**Nucleo-F446RE**

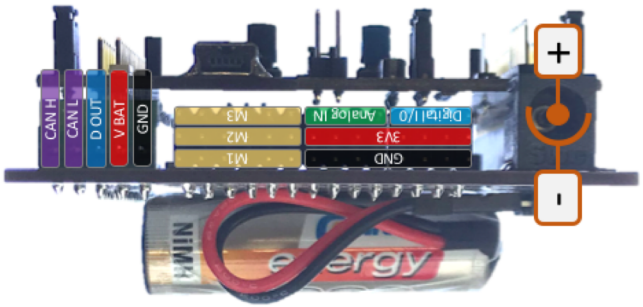
| Nucleo Pin # | STM32 Pin # | Pin Name | CAN | UART | USB | PWM | SPI | I2C | Counter | Analog In | AnalogOut | GPIO | Funktion | Beschreibung | Typ | Level | Bemerkung | Arduino Shield |
|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|---|
| CN7-1 | 51 | PC10 | | UART3_TX / UART4_TX | | | SPI3_SCK | | | | | PC10 | SPI3_SCK | SPI SCK | | | | |
| CN7-2 | 52 | PC11 | | UART3_RX / UART4_RX | | | SPI3_MISO | | | | | PC11 | SPI3_MISO | SPI MISO | | | | |
| CN7-3 | 53 | PC12 | | UART3_CK / UART5_TX | | | SPI3_MOSI | I2C2_SDA | | | | PC12 | SPI3_MOSI | SPI MOSI | | | | |
| CN7-4 | 54 | PD2 | | UART5_RX | | | | | TIM3_ETR | | | PD2 | PD2 | SPI CS XM (IMU) | | | | |
| CN7-5 | 19 | VDD | | | | | | | | | | | | | REF | | | |
| CN7-6 | | E5V | | | | | | | | | | | | | PWR | | | |
| CN7-7 | 60 | BOOT0 | | | | | | | | | | | | | | | | |
| CN7-8 | | GND | | | | | | | | | | | | | GND | | | |
| CN7-9 | | NC | | | | | | | | | | | | | | | NC | |
| CN7-10 | | NC | | | | | | | | | | | | | | | NC | |
| CN7-11 | | NC | | | | | | | | | | | | | | | NC | |
| CN7-12 | | IOREF | | | | | | | | | | | | | REF | | | |
| CN7-13 | 46 | PA13 | | | | | | | | | | PA13 | | SWDIO-JTMS | | 3.3V | SWDIO-JTMS | |
| CN7-14 | 7 | RESET | | | | | | | | | | | | NRST | | | | |
| CN7-15 | 49 | PA14 | | | | | | | | | | PA14 | | SWCLK-JTCK | | 3.3V | SWCLK-JTCK | |
| CN7-16 | | 3V3 | | | | | | | | | | | | | REF | | | |
| CN7-17 | 50 | PA15 | | UART4_RTS | | | SPI1_NSS / SPI3_NSS | | TIM2_CH1 / TIM2_ETR | | | PA15 | PA15 | SPI CS G (IMU) | | 3.3V | JTDI | |
| CN7-18 | | 5V | | | | | | | | | | | | | REF | | | |
| CN7-19 | | GND | | | | | | | | | | | | | GND | | | |
| CN7-20 | | GND | | | | | | | | | | | | | GND | | | |
| CN7-21 | 59 | PB7 | | UART1_RX | | | | I2C1_SDA | TIM4_CH2 | | | PB7 | TIM4_CH2 | Encoder B M2 | | 3.3V | | |
| CN7-22 | | GND | | | | | | §§§ | | | | | | | GND | | | |
| CN7-23 | 2 | PC13 | | | | | | | | | | PC13 | PC13 | Button | | 3.3V | USER_BUTTON | |
| CN7-24 | | VIN | | | | | | | | | | | | | PWR | | | |
| CN7-25 | 3 | PC14 | | | | | | | | | | PC14 | | | | 3.3V | NC(Nucleo) | |
| CN7-26 | | NC | | | | | | | | | | | | | | | NC | |
| CN7-27 | 4 | PC15 | | | | | | | | | | PC15 | | | | 3.3V | NC(Nucleo) | |
| CN7-28 | 14 | PA0 | | UART2_CTS / UART4_TX | | TIM8_ETR | | | TIM2_CH1 / TIM2_ETR / TIM5_CH1 | ADC123_IN0 | | PA0 | TIM2_CH1 | Encoder B M3 | | 3.3V | | Analog In 0 |
| CN7-29 | 5 | PH0 | | | | | | | | | | PH0 | | | | 3.3V | NC(Nucleo) | |
| CN7-30 | 15 | PA1 | | UART2_RTS / UART4_RX | | | | | TIM2_CH2 / TIM5_CH2 | ADC123_IN1 | | PA1 | TIM2_CH2 | Encoder A M3 | | 3.3V | | Analog In 1 |
| CN7-31 | 6 | PH1 | | | | | | | | | | PH1 | PH1 | Motor Feedback | | 3.3V | | |
| CN7-32 | 20 | PA4 | | UART2_CK | | | SPI1_NSS / SPI3_NSS | | | ADC12_IN4 | DAC_OUT1 | PA4 | | | | 3.3V | | Analog In 2 |
| CN7-33 | 1 | VBAT | | | | | | | | | | | | | | | | |
| CN7-34 | 26 | PB0 | | UART4_CTS | | TIM1_CH2N / TIM8_CH2N | SPI3_MOSI | | TIM3_CH3 | ADC12_IN8 | | PB0 | ADC12_IN8 | | | 3.3V | | Analog In 3 |
| CN7-35 | 10 | PC2 | | | | | SPI2_MISO | | | ADC123_IN12 | | PC2 | ADC123_IN12 | Analog0 | | 3.3V | | |
| CN7-36 | 9 | PC1 | | | | | SPI2_MOSI / SPI3_MOSI | | | ADC123_IN11 | | PC1 | PC1 | | | 3.3V | NC(Nucleo) | Analog In 4 |
| CN7-37 | 11 | PC3 | | | | | SPI2_MOSI | | | ADC123_IN13 | | PC3 | ADC123_IN13 | Analog1 | | 3.3V | | |
| CN7-38 | 8 | PC0 | | | | | | | | ADC123_IN10 | | PC0 | PC0 | | | 3.3V | | Analog In 5 |
| CN10-1 | 40 | PC9 | | UART5_CTS | | TIM8_CH4 | | I2C3_SDA | TIM3_CH4 | | | PC9 | PC9 | Servo 0 / D0 | | 3.3V | | |
| CN10-2 | 39 | PC8 | | UART5_RTS / UART6_CK | | TIM8_CH3 | | | TIM3_CH3 | | | PC8 | PC8 | Servo 1 / D1 | | 3.3V | | |
| CN10-3 | 61 | PB8 | CAN1_RX | | | | | I2C1_SCL | TIM2_CH1 / TIM2_ETR / TIM4_CH3 / TIM10_CH1 | | | PB8 | I2C1_SCL | | | 3.3V | | SCL |
| CN10-4 | 37 | PC6 | | UART6_TX | | TIM8_CH1 | | | TIM3_CH1 | | | PC6 | PC6 | Servo 2 / D2 | | 3.3V | | |
| CN10-5 | 62 | PB9 | CAN1_TX | | | | SPI2_NSS | I2C1_SDA | TIM2_CH2 / TIM4_CH4 / TIM11_CH1 | | | PB9 | I2C1_SDA | | | 3.3V | | SDA |
| CN10-6 | 25 | PC5 | | UART3_RX | | | | | | ADC12_IN15 | | PC5 | ADC12_IN15 | Analog2 | | 3.3V | | |
| CN10-7 | 13 | AVDD | | | | | | | | | | | | | REF | | | AREF |
| CN10-8 | | U5V | | | | | | | | | | | | | REF | | | |
| CN10-9 | | GND | | | | | | | | | | | | | GND | | | GND |
| CN10-10 | | NC | | | | | | | | | | | | | | | NC | |
| CN10-11 | 21 | PA5 | | | | TIM8_CH1N | SPI1_SCK | | TIM2_CH1 / TIM2_ETR | ADC12_IN5 | DAC_OUT2 | PA5 | PA5 | | | 3.3V | LED | SCK |
| CN10-12 | 45 | PA12 | CAN1_TX | UART1_RTS | USB_FS_D+ | TIM1_ETR | | | | | | PA12 | CAN1_TX | CAN TX | | 3.3V | | |
| CN10-13 | 22 | PA6 | | | | TIM1_BKIN / TIM8_BKIN | SPI1_MISO | | TIM3_CH1 / TIM13_CH1 | ADC12_IN6 | | PA6 | TIM3_CH1 | Encoder A M1 | | 3.3V | | MISO |
| CN10-14 | 44 | PA11 | CAN1_RX | UART1_CTS | USB_FS_D- | TIM1_CH4 | | | | | | PA11 | CAN1_RX | CAN RX | | 3.3V | | |
| CN10-15 | 23 | PA7 | | | | TIM1_CH1N / TIM8_CH1N | SPI1_MOSI | | TIM3_CH2 / TIM14_CH1 | ADC12_IN7 | | PA7 | PA7 | | | 3.3V | | MOSI |
| CN10-16 | 33 | PB12 | CAN2_RX | UART3_CK | | TIM1_BKIN | SPI2_NSS | I2C2_SMBA | | | | PB12 | PB12 | Servo 3 / D3 | | 3.3V | | |
| CN10-17 | 58 | PB6 | CAN2_TX | UART1_TX | | | | I2C1_SCL | TIM4_CH1 | | | PB6 | TIM4_CH1 | Encoder A M2 | | 3.3V | | CS |
| CN10-18 | | NC | | | | | | | | | | | | | | | | |
| CN10-19 | 38 | PC7 | | UART6_RX | | TIM8_CH2 | SPI2_SCK | | TIM3_CH2 | | | PC7 | TIM3_CH2 | Encoder B M1 | | 3.3V | | PWM |
| CN10-20 | | GND | | | | | | | | | | | | | GND | | | |
| CN10-21 | 42 | PA9 | | UART1_TX | | TIM1_CH2 | SPI2_SCK | I2C3_SMBA | | | | PA9 | TIM1_CH2 | PWM Motor M2 | | 3.3V | | GPIO |
| CN10-22 | 28 | PB2 | | | | | SPI3_MOSI | | TIM2_CH4 | | | PB2 | PB2 | Enable Motor M2 | | 3.3V | | |
| CN10-23 | 41 | PA8 | | UART1_CK | | TIM1_CH1 | | I2C3_SCL | | | | PA8 | TIM1_CH1 | PWM Motor M1 | | 3.3V | | GPIO |
| CN10-24 | 27 | PB1 | | | | TIM1_CH3N / TIM8_CH3N | | | TIM3_CH4 | ADC12_IN9 | | PB1 | ADC12_IN9 | Analog3 | | 3.3V | | |
| CN10-25 | 29 | PB10 | | UART3_TX | | | SPI2_SCK | I2C2_SCL | TIM2_CH3 | | | PB10 | UART3_TX | | | 3.3V | | PWM |
| CN10-26 | 36 | PB15 | | | USB_HS_D+ | TIM1_CH3N / TIM8_CH3N | SPI2_MOSI | | TIM12_CH2 | | | PB15 | TIM12_CH2 | | | 3.3V | | |
| CN10-27 | 56 | PB4 | | | | | SPI1_MISO / SPI3_MISO / SPI2_NSS | I2C3_SDA | TIM3_CH1 | | | PB4 | | | | 3.3V | NJTRST | PWM |
| CN10-28 | 35 | PB14 | | UART3_RTS | USB_HS_D- | TIM1_CH2N / TIM8_CH2N | SPI2_MISO | | TIM12_CH1 | | | PB14 | TIM12_CH1 | | | 3.3V | | |
| CN10-29 | 57 | PB5 | CAN2_RX | | | | SPI1_MOSI / SPI3_MOSI | I2C1_SMBA | TIM3_CH2 | | | PB5 | | | | 3.3V | | GPIO |
| CN10-30 | 34 | PB13 | CAN2_TX | UART3_CTS | | TIM1_CH1N | SPI2_SCK | | | | | PB13 | PB13 | Enable Motor | | 3.3V | | |
| CN10-31 | 55 | PB3 | | | | | SPI1_SCK / SPI3_SCK | I2C2_SDA | TIM2_CH2 | | | PB3 | | | | 3.3V | JTDO | PWM |
| CN10-32 | | AGND | | | | | | | | | | | | | AGND | | | |
| CN10-33 | 43 | PA10 | | UART1_RX | | TIM1_CH3 | | | | | | PA10 | TIM1_CH3 | PWM M3 | | 3.3V | | GPIO |
| CN10-34 | 24 | PC4 | | | | | | | | ADC12_IN14 | | PC4 | PC4 | PWR Out 1 | | 3.3V | | |
| CN10-35 | 16 | PA2 | | UART2_TX | | | | | TIM2_CH3 / TIM5_CH3 / TIM9_CH1 | ADC123_IN2 | | PA2 | UART2_TX | ST Link Serial Port | | 3.3V | COM-PORT | Serial TX |
| CN10-36 | | NC | | | | | | | | | | | | | | | NC | |
| CN10-37 | 17 | PA3 | | UART2_RX | | | | | TIM2_CH4 / TIM5_CH4 / TIM9_CH2 | ADC123_IN3 | | PA3 | UART2_RX | ST Link Serial Port | | 3.3V | COM-PORT | Serial RX |
| CN10-38 | | NC | | | | | | | | | | | | | | | NC | |

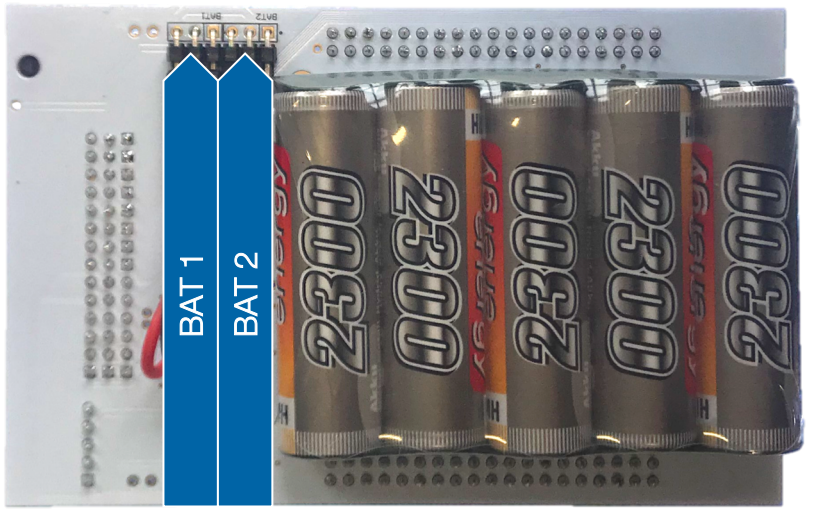
# PES 2 Board - Leistung



# PES 2 Board - Ladungsanzeige

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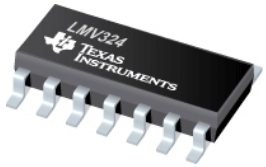
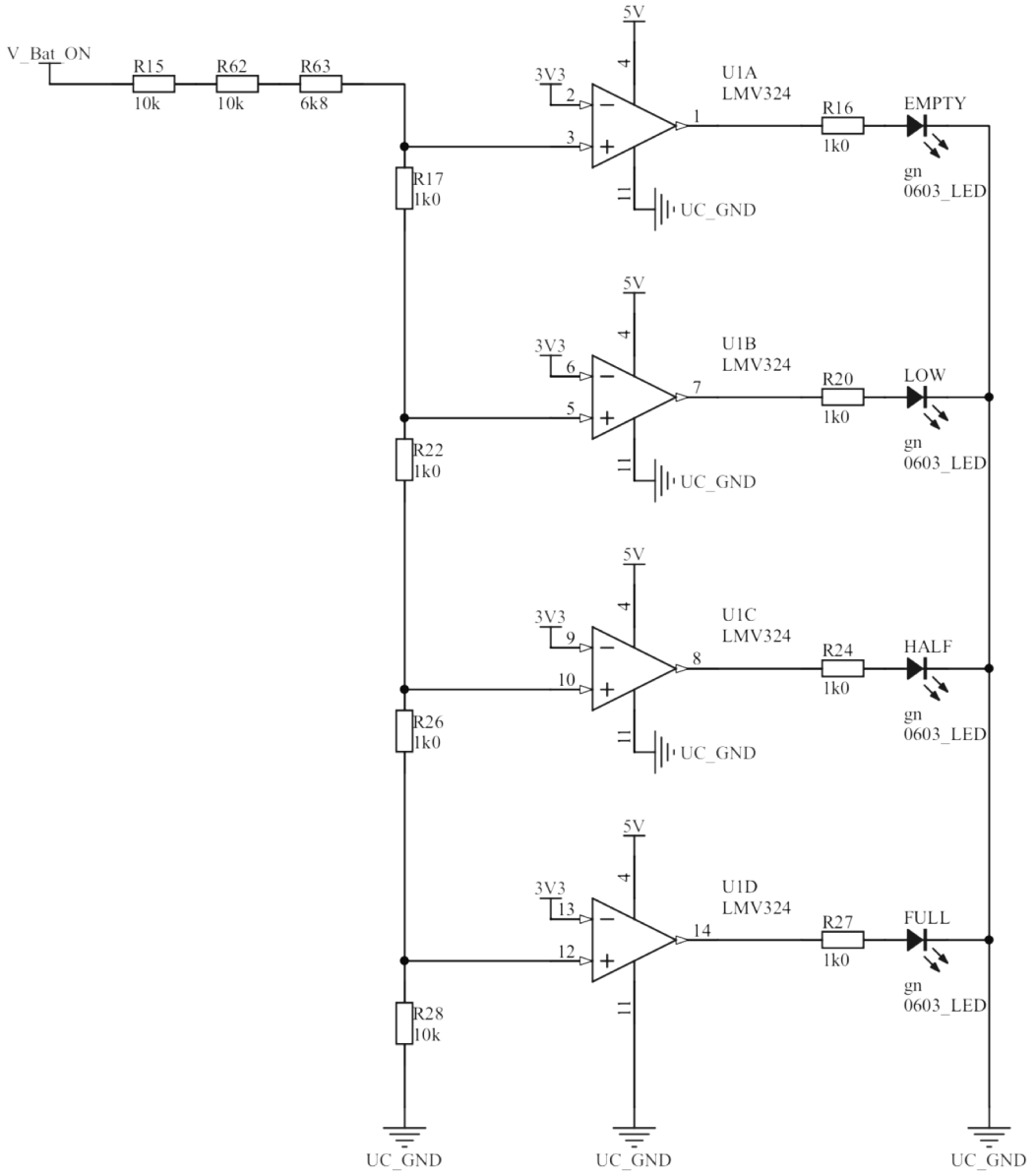
## LMV 324D




Quad Operational Amplifier

- Full: 13.1V
- Half: 11.9V
- Low: 10.9V
- Empty: 10.1V



# Der Antrieb - Treiber IC

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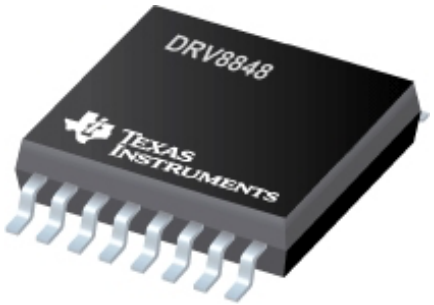
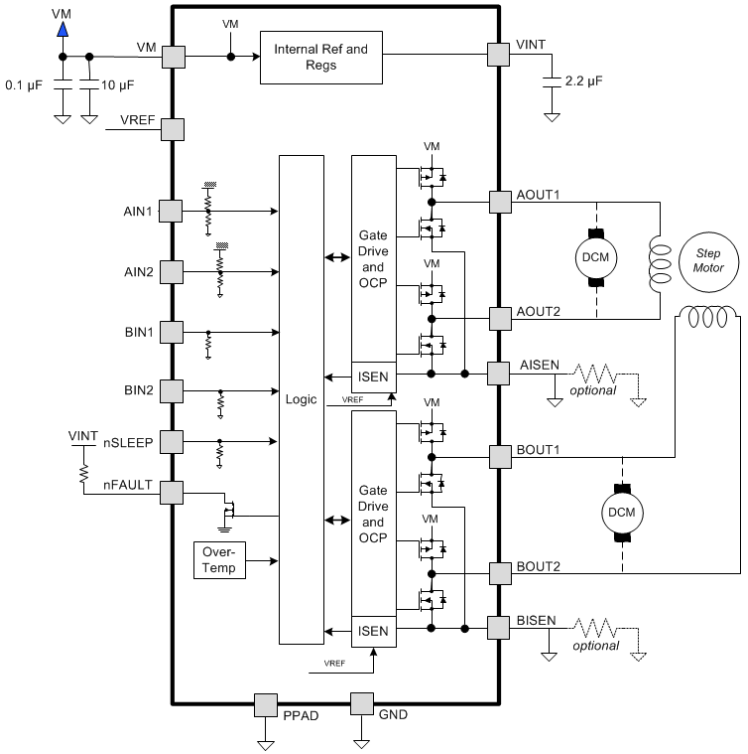
## Treiber: TI DRV 8848

2A Dual H-Bridge Motor Driver (PWM Control)




- 2x Vollbrückentreiber
- Betriebsspannung: 4 – 18V
- 2x 1A Nennstrom



# Der Antrieb - Vollbrückenschaltung

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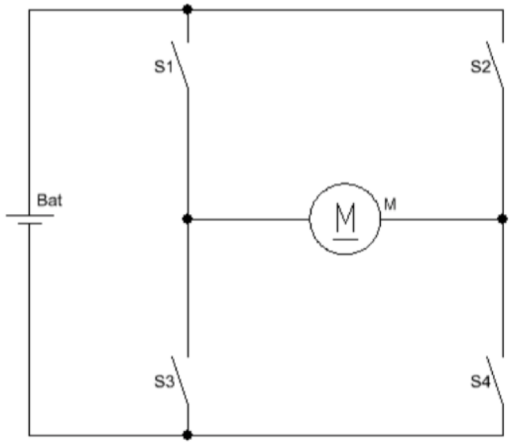


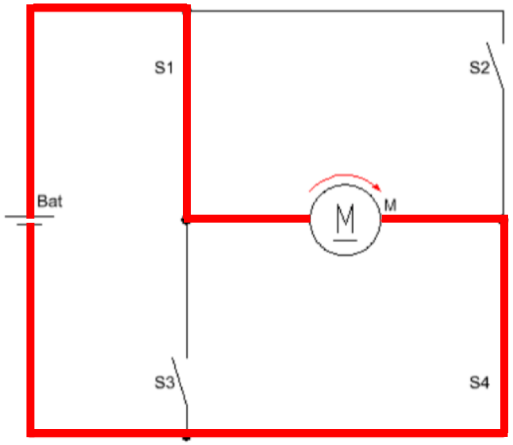


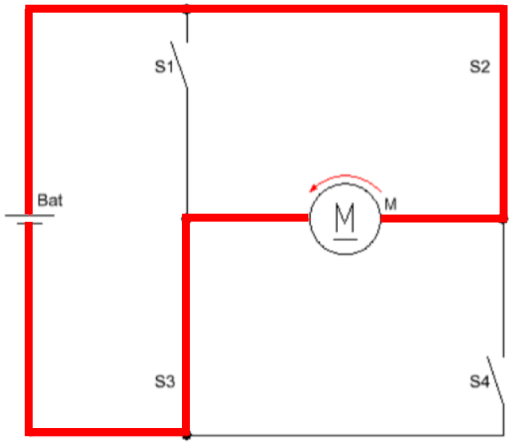


**Stillstand:**
S1 = 0 | S2 = 0
S3 = 0 | S4 = 0

**Rechtsläufig:**
S1 = 1 | S2 = 0
S3 = 0 | S4 = 1

**Linksläufig:**
S1 = 0 | S2 = 1
S3 = 1 | S4 = 0

# Der Antrieb - Ansteuerung

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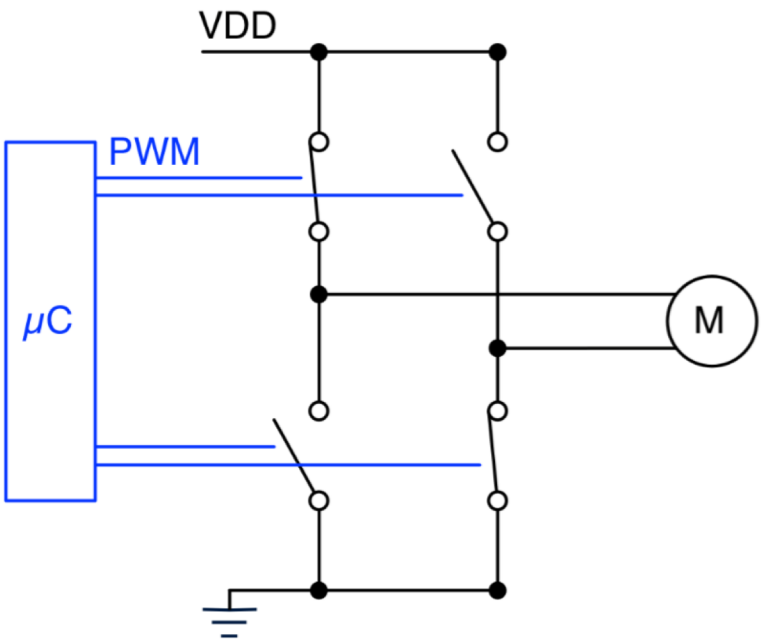
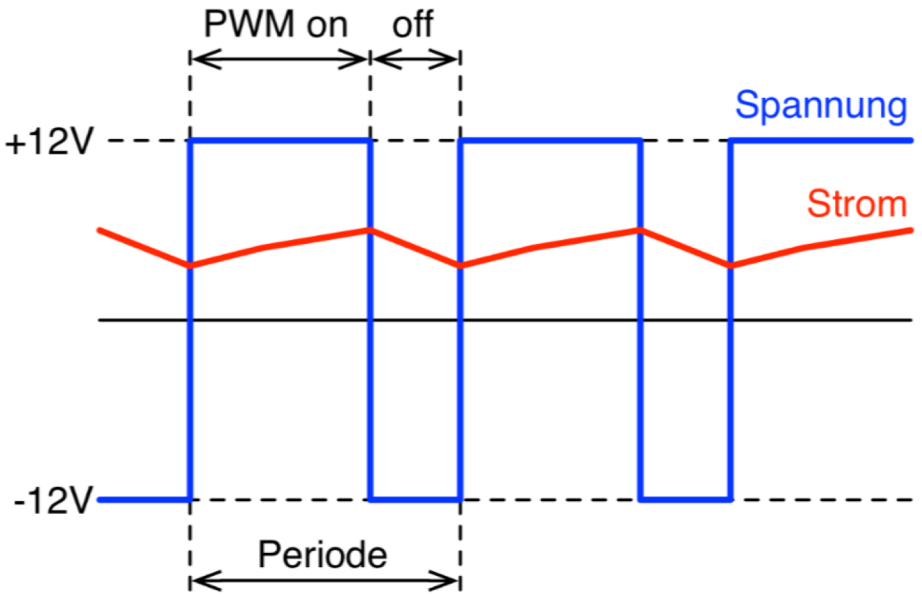
- Ansteuerung mit PWM Signal des Mikrokontrollers:
  - Die Phasenspannung entspricht abwechslungsweise der positiven oder negativen Versorgungsspannung



# Der Antrieb - Pulsweiten Modulation

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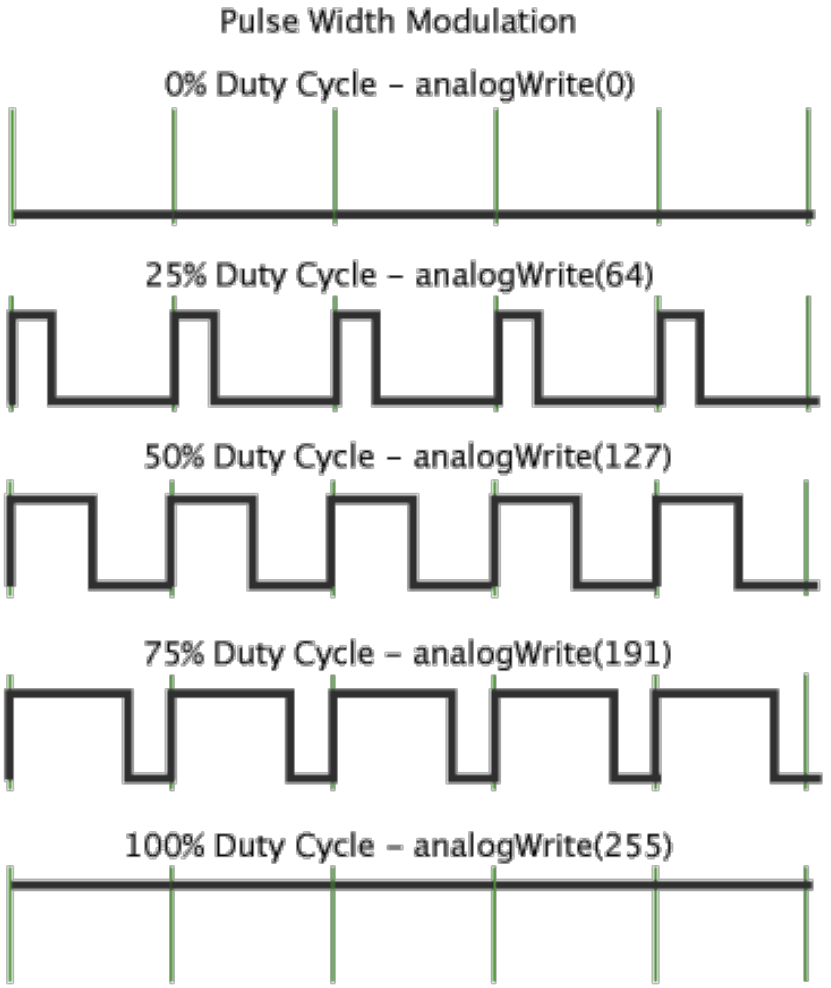
Das Signal wiederholt sich periodisch.
Das Tastverhältnis (duty cycle) wird für jeden Motor separat angegeben



# Der Antrieb - Pulsweiten Modulation

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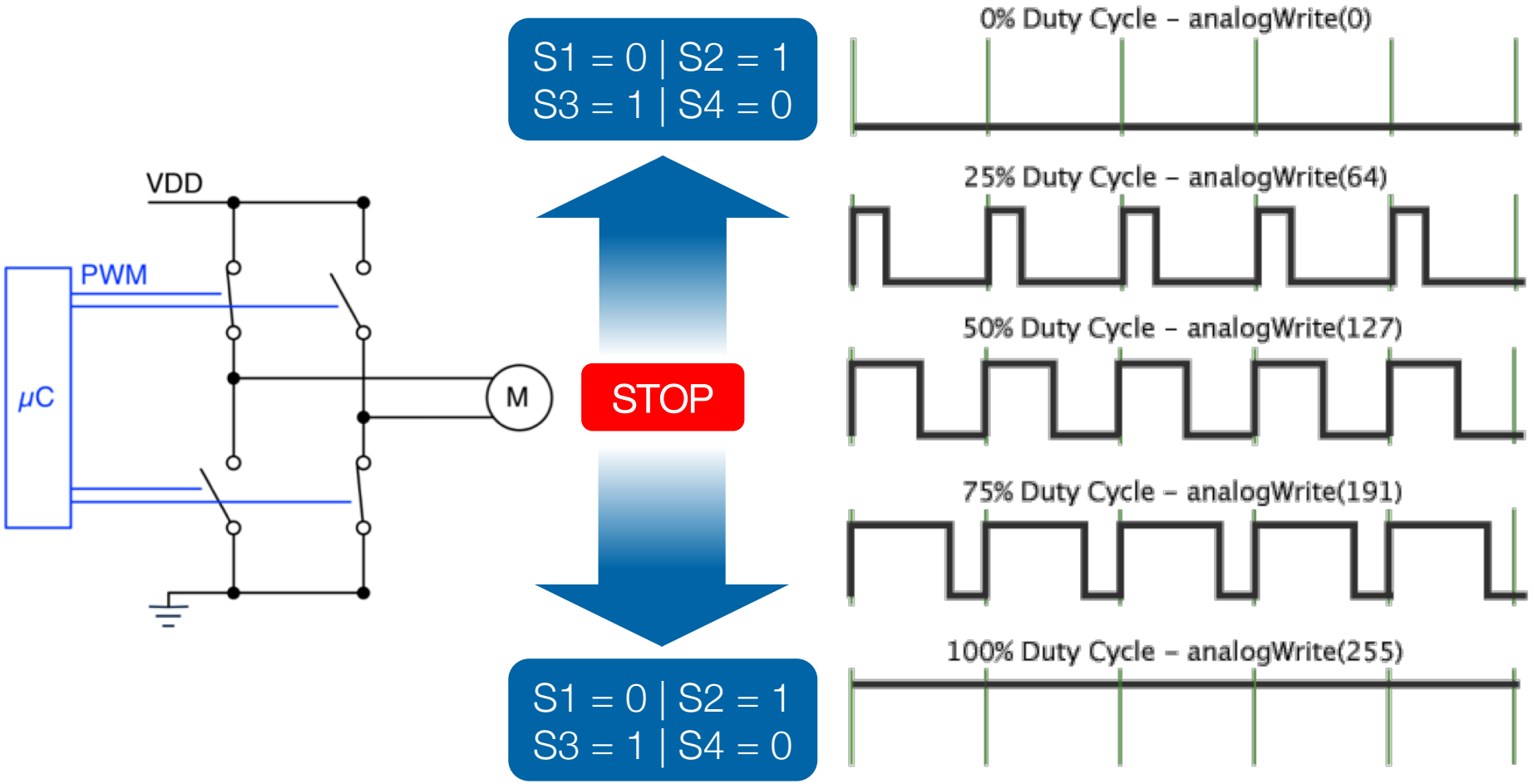
## Komplementäre Ansteuerung von S1/S4 und S2/S3



# Der Antrieb - Schaltung

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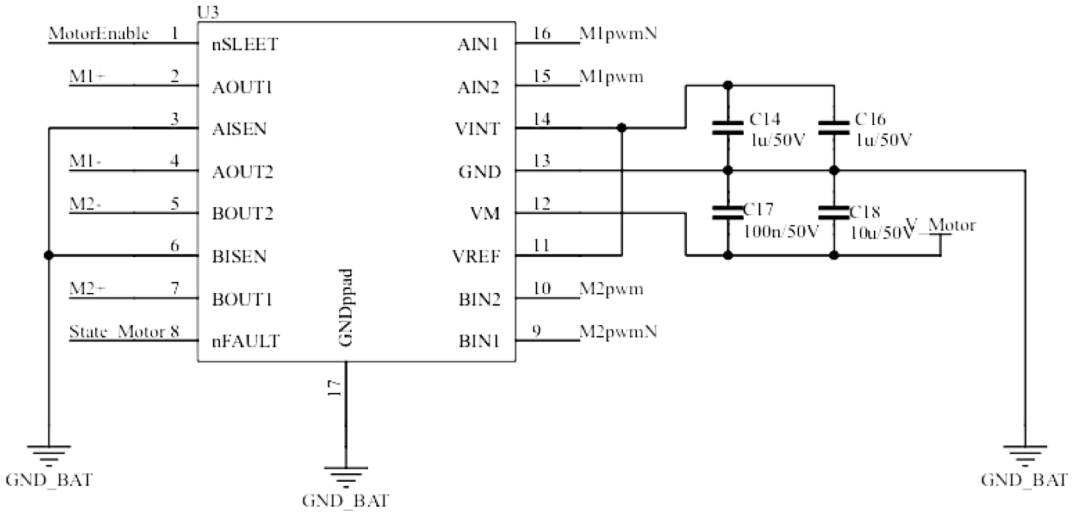
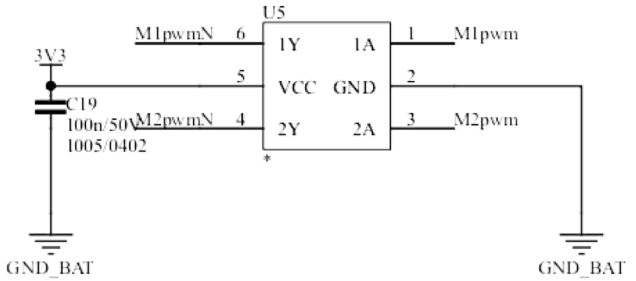
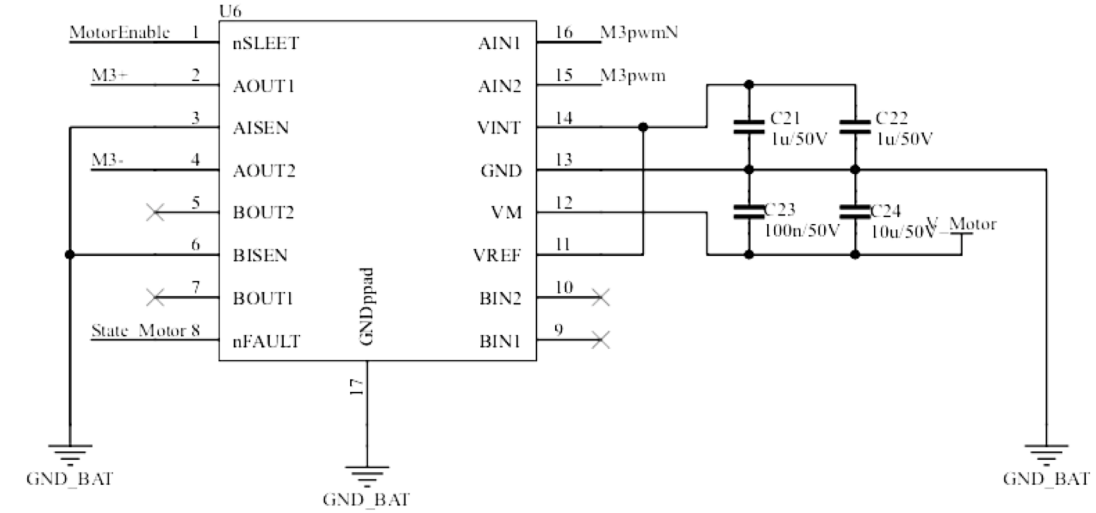
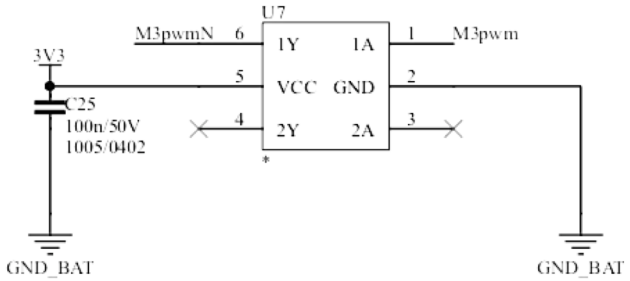


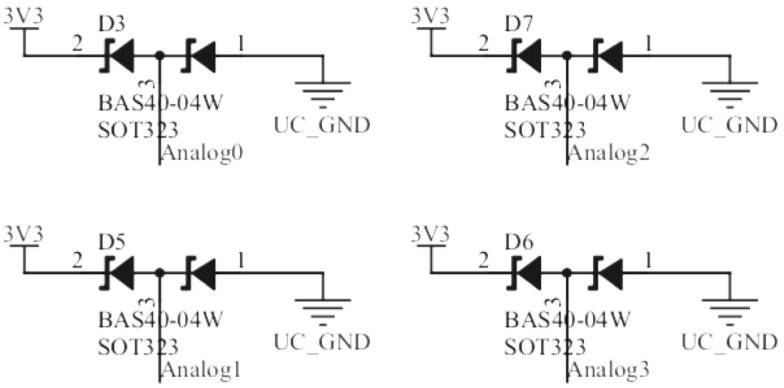



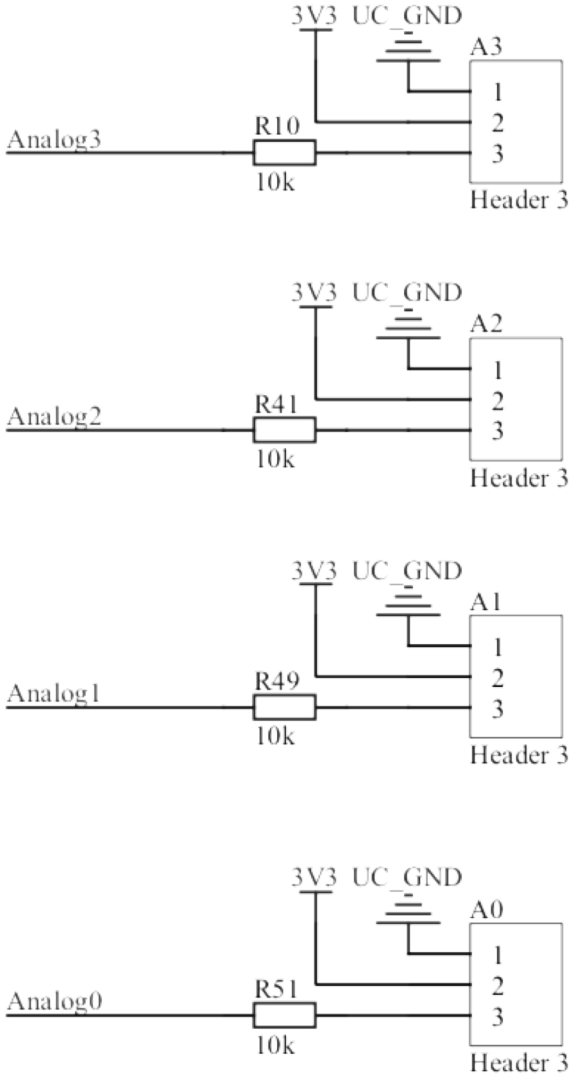



# Analog IN

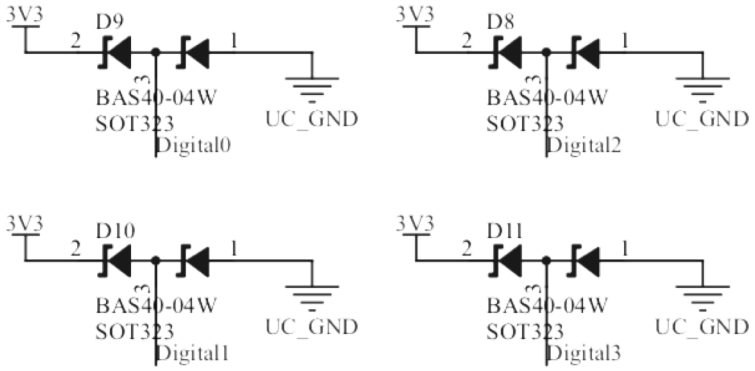
- ESD Schutzdioden:
  *Überspannungsschutz*
- 10k Eingangswiderstand:
  *Strombegrenzung*

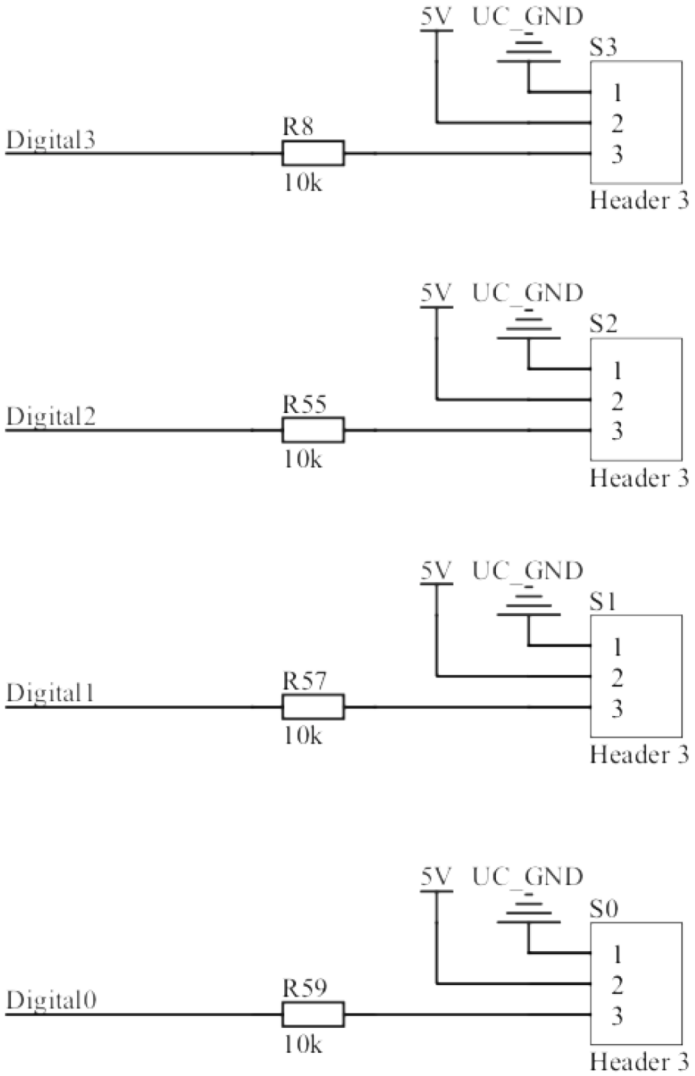


# Digital I/O

- ESD Schutzdioden:
  *Überspannungsschutz*
- 10k Eingangswiderstand:
  *Strombegrenzung*



# Schutzschaltung - Detail

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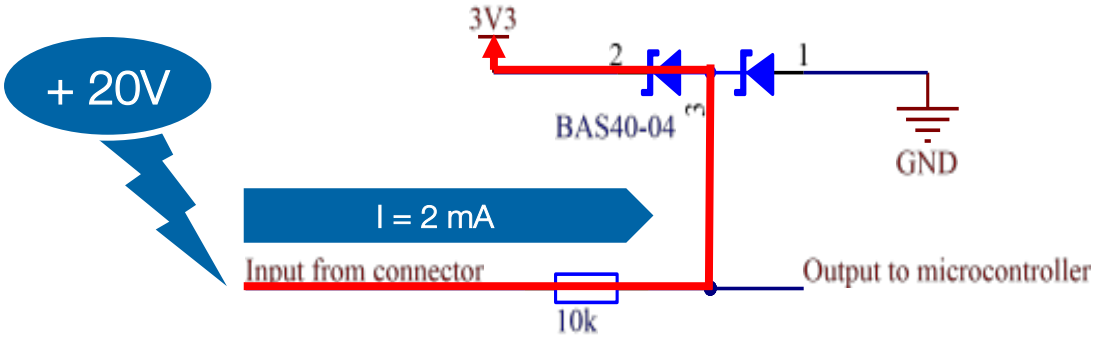
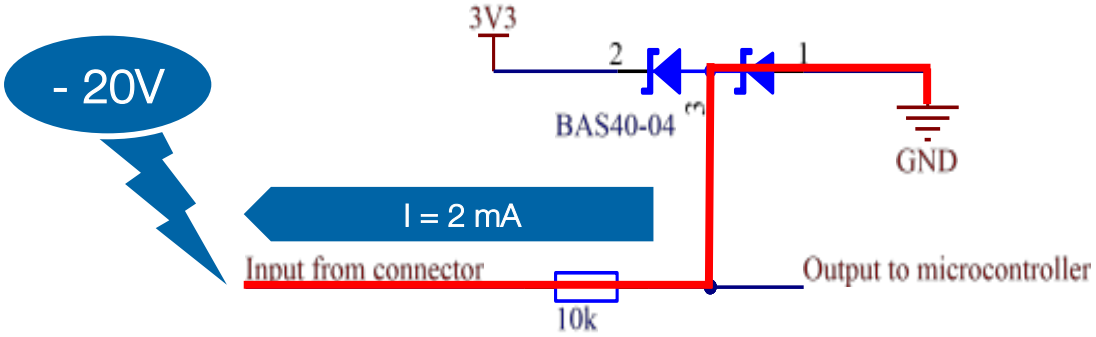
- Überspannung



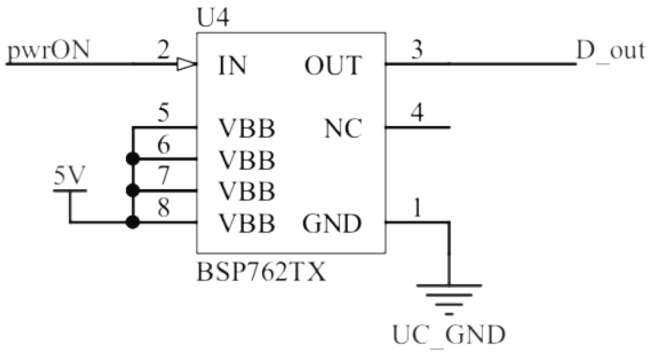

- Verpolung



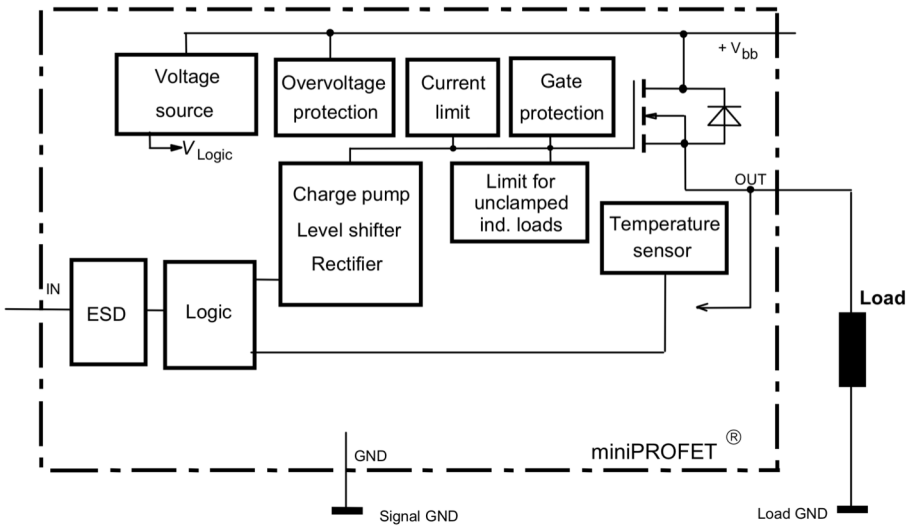
# High Side Switch

## Treiber: Infineon BSP 762 T
Smart Power High Side Switch

P-DSO-8




- Treiber für ohmsche, induktive und kapazitive Lasten
- Betriebsspannung: 5 – 34V
- 2A Nennstrom
  → *Auf PES Board max. 1A!!!*



# IMU

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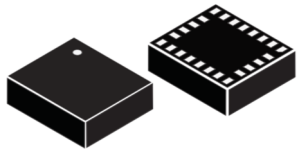
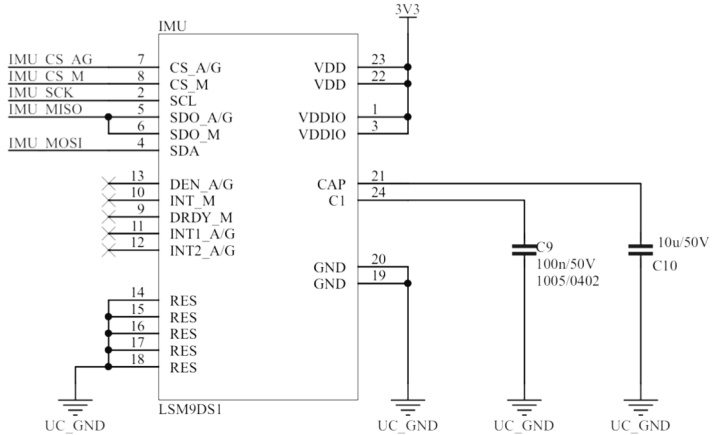
## 9-DOF IMU: ST LSM9DS1

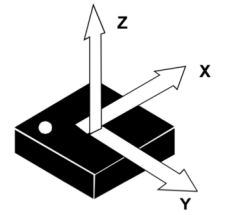
MEMS Sensor



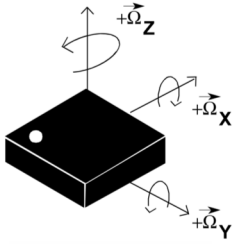
LGA-24L (3.5x3x1.0 mm)



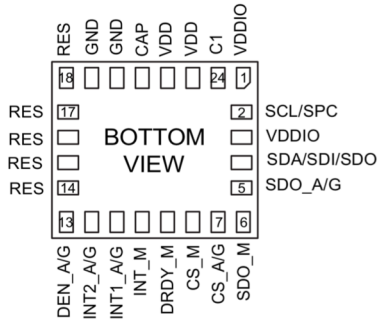

- 3-Achsen Beschleunigung
- 3-Achsen Gyro (Drehrate)
- 3-Achsen Kompass
- Schnittstelle: SPI




(TOP VIEW)
DIRECTIONS OF THE DETECTABLE ACCELERATIONS




(TOP VIEW)
DIRECTIONS OF THE DETECTABLE ANGULAR RATES



(TOP VIEW)
DIRECTIONS OF THE DETECTABLE MAGNETIC FIELDS

# Euer PES 2 Set – Viel Spass!

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# DC Getriebemotor mit Encoder

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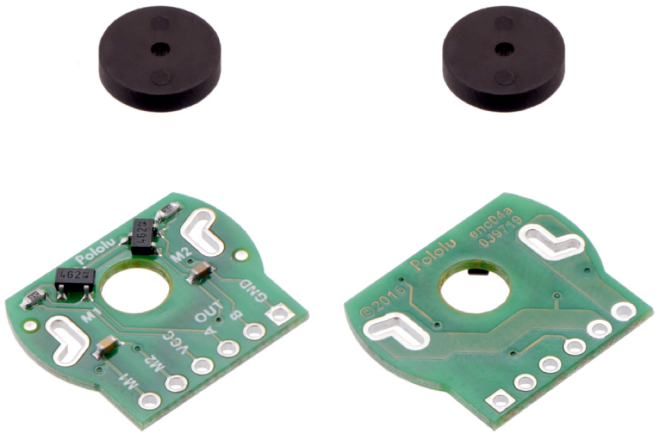
IMS Institute of Mechatronic Systems

- 20D mm Metal Gearmotors 12V
  *Preis ca. 23 $ (Stück)*

  → *ACHTUNG! Dual-Shaft Version wählen!*

| Rated Voltage | Brush Type | Stall Current @ Rated Voltage | No-Load Speed @ Rated Voltage | Approximate Stall Torque @ Rated Voltage | Pololu Single-Shaft (Gearbox Only) | Pololu Dual-Shaft (Gearbox & Motor) |
|---|---|---|---|---|---|---|
| 12 V | long-life carbon (CB) | 1.6 A | 570 RPM | 26 oz-in | 25:1 12V CB | 25:1 12V CB dual shaft |
| | | | 450 RPM | 33 oz-in | 31:1 12V CB | 31:1 12V CB dual shaft |
| | | | 225 RPM | 58 oz-in | 63:1 12V CB | 63:1 12V CB dual shaft |
| | | | 180 RPM | 75 oz-in | 78:1 12V CB | 78:1 12V CB dual shaft |
| | | | 140 RPM | 85 oz-in | 100:1 12V CB | 100:1 12V CB dual shaft |
| | | | 110 RPM | 110 oz-in | 125:1 12V CB | 125:1 12V CB dual shaft |
| | | | 90 RPM | 130 oz-in | 156:1 12V CB | 156:1 12V CB dual shaft |
| | | | 72 RPM | 160 oz-in | 195:1 12V CB | 195:1 12V CB dual shaft |
| | | | 57 RPM | 200 oz-in | 250:1 12V CB | 250:1 12V CB dual shaft |
| | | | 45 RPM | 250 oz-in | 313:1 12V CB | 313:1 12V CB dual shaft |
| | | | 36 RPM | 290 oz-in | 391:1 12V CB | 391:1 12V CB dual shaft |
| | | | 29 RPM | 350 oz-in | 488:1 12V CB | 488:1 12V CB dual shaft |

- Magnetic Encoder Pair Kit for 20D mm Metal Gearmotors, 20 CPR, 2.7-18V
  *Preis ca. 9 $ (Paar)*




- Erhältlich bei [www.pololu.com](http://www.pololu.com)

# Sensorik – Anbieter und Info

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- https://www.pololu.com
- https://www.adafruit.com
- https://www.sparkfun.com
- https://www.seeedstudio.com
- https://www.robotshop.com
- https://boxtec.ch
- https://www.play-zone.ch
- http://farnell.ch
- https://www.mouser.ch
- https://www.digikey.com
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RobotShop Putting robotics at your service™

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